
a
Five-group ancestry classification
hEVO-large
Evo2-7B
NTv3-650M
NTv1-500M-1kG
DNABERT
DNABERT2
GPN-MSA
GPN-Star-V
GENA-LM-large
GENERator-3B
NTv2-500M-MS
AIDO.DNA-300M
Mistral-DNA-422M
GPN-Promoter
Genos-1.2B
GROVER
hEVO-base
Caduceus-PS
SpeciesLM
Caduceus-PH
HyenaDNA-1K
0.94
0.96
0.98
MCC
b
Few-shot ancestry-proportion prediction
60 samples per population
Mean Pearson r
0.96
0.88
0.80
0.72
10
20
30
40
50
60
Training samples per population
hEVO-large
Evo2-7B
NTv1-2.5B-1kG
GENERator-3B
NTv3-650M
GPN-MSA
GPN-Star-P
DNABERT
NTv2-500M-MS
GENA-LM-base
AIDO.DNA-300M
Genos-1.2B
DNABERT2
Mistral-DNA-422M
hEVO-base
GPN-Promoter
GROVER
SpeciesLM
Caduceus-PS
Caduceus-PH
HyenaDNA-1K
0.86
0.90
0.94
Mean Pearson r
c
HAR
hEVO-large
GPN-Star-M
GPN-MSA
hEVO-base
GENA-LM-large
Evo2-7B
DNABERT2
Caduceus-PS
NTv1-2.5B-1kG
DNABERT
Caduceus-PH
Genos-1.2B
Mistral-DNA-422M
NTv3-650M
SpeciesLM
GROVER
AIDO.DNA-300M
GENERator-3B
NTv2-500M-MS
GPN-Promoter
HyenaDNA-1K
0.871
0.813
0.2
0.4
0.6
0.8
0.9
MCC
d
HAQER
hEVO-large
GPN-Star-M
hEVO-base
GENA-LM-large
Evo2-7B
Genos-1.2B
DNABERT2
DNABERT
NTv1-2.5B-1kG
Mistral-DNA-422M
GROVER
NTv3-650M
GPN-MSA
GPN-Promoter
AIDO.DNA-300M
SpeciesLM
NTv2-500M-MS
Caduceus-PS
HyenaDNA-1K
GENERator-1.2B
Caduceus-PH
0.843
0.684
0.2
0.4
0.6
0.8
0.9
MCC
hEVO
Evo 2
NT
GPN
Other
e
1 kb
hEVO-large score
0.59
GPN-Star-M score
0.59
hEVO calls
Known HAQER
Ape divergence
ENCODE cCREs
RefSeq genes
ITSN1 →
COL4A2 →
C17orf67 ←
C22orf34 ←
CDH13 →
21:33.896
20:61.218
13:110.314
17:56.793
18:24.841
22:49.596
13:34.909
22:48.044
16:83.511
15:75.240
16:89.069
20:23.156
hg38 midpoint (Mb); ticks: -2, 0, +2 kb; // separate loci
f
Adaptive-allele benchmark (30 variants)
1 Europe (6)
LCT/MCM6 · SLC45A2 · TYR
HERC2/OCA2 · IRF4 · KITLG
Milk digestion · pigmentation
2 Africa / Arabia (2)
LCT/MCM6
Lactase persistence
3 Sub-Saharan Africa (4)
HBB · ACKR1 · SLC39A4
Malaria protection · zinc
4 Tibetan Plateau (3)
EGLN1 · EPAS1
High-altitude hypoxia
5 East Asia (6)
OCA2 · MC1R · EDAR
ABCC11 · ALDH2
Pigmentation · hair / glands
Earwax · alcohol metabolism
6 Andean highlands (1)
PRKAA1
Hypoxic ventilation
7 N. Eurasia (1)
TRPM8
Cold perception
8 W. Eurasia (2)
FADS1/FADS2 · TLR1
Lipids · innate immunity
B Broad distributions (5)
SLC24A5 · KITLG · OAS1
SLC30A9 · CASP12
Pigmentation · zinc · immunity
hEVO-base
hEVO-large
Mistral-DNA-17M
GENA-LM-base
NTv2-500M-MS
Caduceus-PS
HyenaDNA-1K
AIDO.DNA-300M
Caduceus-PH
GROVER
NTv1-2.5B-MS
NTv3-100M
SpeciesLM
DNABERT2
Evo2-1B
GENERator-1.2B
GPN-Promoter
Genos-1.2B
GPN-Star-P
GPN-MSA
Selected-allele preference
0
50
100
Loci (%)
Calinski-Harabasz index
10⁻⁴ 10⁻² 1
Raw scores, log scale